



ARIN PUBLISHING ACADEMY

Research Writing, Publishing, and Policy Communication for African Scholars

MODULE 1: ACADEMIC WRITING MASTERY

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Session 2:

Building a Strong Manuscript



Topics

01

Writing effective abstracts

02

Literature review development

03

Constructing arguments

04

Presenting methods and results

Recap...

- A strong manuscript is more than a collection of research findings.
- It tells a clear scientific story that begins with an important research question, demonstrates rigorous methods, presents evidence objectively, and concludes with meaningful implications.
- Every section of the manuscript has a unique purpose, yet all sections must work together to guide readers through the research.

A man with dark hair, wearing a blue shirt, is shown from the chest up, looking slightly to the right with a thoughtful expression. The background is blurred, showing what appears to be a store or office setting with shelves and lights. Overlaid on the image are three white rectangular boxes containing text. The first box, at the top, contains the text 'Abstract vs. Introduction:' in red. The second box, in the middle, contains the text 'Do You Know the' in black. The third box, at the bottom, contains the text 'Difference?' in black. The entire image is framed by a grey border, and there is an orange triangle in the bottom right corner of the overall slide.

Abstract vs. Introduction:

**Do You Know the
Difference?**

Abstract vs Introduction

An **abstract** and an **introduction** serve different purposes, even though they both appear at the beginning of a document. The key difference is that an **abstract summarizes the entire document**, while an **introduction sets the stage for the discussion that follows**.

Abstract	Introduction
Provides a concise summary of the entire document.	Introduces the topic and explains why it is important.
Enables readers to quickly determine whether the document is relevant to them.	Provides background and context to help readers understand the issue.
Summarizes the problem, purpose, methods (if applicable), key findings or messages, and conclusions or recommendations.	Describes the problem, context, rationale and objectives, but does not summarize the entire document.
Written after the document is completed, although it appears first.	Written at the beginning of the document development process and refined as needed.
Usually one paragraph (150–300 words, depending on the document).	Can range from one paragraph to several pages, depending on the document type.
Should be able to stand alone without reading the rest of the document.	Leads readers into the main body of the document and relies on subsequent sections for details.

Learning Objectives

By the end of this session, participants will be able to:

- Write a structured, concise, and compelling abstract.
- Develop a literature review that synthesizes evidence and establishes the research gap.
- Construct logical, evidence-based arguments throughout a manuscript.
- Present methods and results clearly, transparently, and according to international reporting standards.
- Integrate the different manuscript sections into a coherent scientific paper.

Writing Effective Abstracts

Why the Abstract Matters

- The abstract is the most frequently read section of any manuscript.
- Editors often make their initial decision based on the abstract.
- Researchers use abstracts to determine whether they should read the full article.
- Search engines index abstracts, making them essential for discoverability.
- A weak abstract can cause an excellent study to be overlooked.

Characteristics of an Effective Abstract

An effective abstract is:

- Accurate
- Concise
- Complete
- Clear
- Self-contained
- Consistent with the manuscript

It should never introduce information that is absent from the main paper.

STRUCTURING THE ABSTRACT

A clear, structured abstract helps readers quickly understand your study.

BACKGROUND	Briefly explain the problem.	Example: <i>Depression affects one in four adolescents in Kenya, yet evidence on scalable school-based interventions remains limited.</i>
OBJECTIVE	Clearly state what the study sought to investigate.	Example: <i>This study evaluated the effectiveness of the Shamiri Intervention in reducing depressive symptoms among secondary school students.</i>
METHODS	Include • Study design • Setting • Participants • Sample size • Main analysis	Example: <i>We conducted a cluster randomized controlled trial involving 1,200 adolescents from 20 public secondary schools.</i>
RESULTS	Present only the most important findings. Include: • Effect sizes • Confidence intervals • p-values when appropriate	Example: <i>Participants receiving the intervention demonstrated significantly lower depressive symptoms than controls ($\beta = -2.4$, $p < 0.01$).</i>
CONCLUSION	State the implications.	Example: <i>School-based interventions may provide an effective and scalable strategy for improving adolescent mental health in low-resource settings.</i>

COMMON MISTAKES IN ABSTRACT WRITING



Avoid these pitfalls to write a *clear*, *accurate*, and *impactful* abstract.



1

Writing the abstract before finishing the manuscript

This can lead to inaccurate, incomplete, or inconsistent information.



2

Including citations

Abstracts should stand alone and do not require references.



3

Including unnecessary background

Keep background information minimal and relevant to the study.



4

Reporting results not found in the paper

The abstract must accurately reflect the study's actual findings.



5

Using vague statements

Avoid general or unclear language. Be specific and precise.



6

Exceeding word limits

Follow the target journal's word limit strictly.



KEY TAKEAWAY:

A well-written abstract is *concise*, *complete*, and *accurate*—giving readers the right picture of your research.



Practical Exercise: Improving a Poorly Written Abstract

Instructions

Read the abstract on next slide and. Identify **at least five areas** that could be improved before submission to a journal.

Consider:

- Is all the essential information included?
- Is the writing clear and concise?
- Are the results reported effectively?
- Does the conclusion follow from the findings?

Antimicrobial Resistance Among Patients with Urinary Tract Infections in County Referral Hospitals in Kenya

Abstract

Urinary Tract Infections (UTIs) are among the most common bacterial infections treated in healthcare facilities, and increasing antimicrobial resistance threatens the effectiveness of commonly prescribed antibiotics. This study assessed antimicrobial resistance patterns among bacterial isolates from patients diagnosed with UTIs in selected county referral hospitals in Kenya. A cross-sectional study was conducted in four county referral hospitals, where urine samples from patients with suspected UTIs were cultured and antimicrobial susceptibility testing was performed using standard laboratory procedures. Data were analysed using descriptive statistics and chi-square tests. A total of 642 bacterial isolates were analysed, with *Escherichia coli* identified as the most frequently isolated organism. High resistance was observed to ampicillin and trimethoprim-sulfamethoxazole, while lower resistance was found for nitrofurantoin. Resistance patterns also differed across hospitals. The findings suggest that antimicrobial resistance remains an important public health concern, and continued surveillance is needed to support evidence-based treatment guidelines.

Literature Review Development

A literature review is more than a summary of previous studies. Its purpose is to

1. Critically examine existing knowledge,
2. Identify what is already established,
3. Highlight remaining gaps
4. Demonstrate how the current study will contribute to the field.

By the end of the literature review, readers should clearly see the gap and understand why the research is both relevant and necessary.

The Purpose of a Literature Review

A literature review should answer four key questions that justify the need for your study:

1. **What do we already know?** Summarize the current evidence, highlighting the major findings, established knowledge, and areas of consensus from previous research.
2. **What do we still not know?** Identify the gaps, inconsistencies, limitations, or unanswered questions that remain in the existing body of literature.
3. **Why is this study necessary?** Explain why addressing the identified gap is important and how the findings could contribute to research, policy, clinical practice, or decision-making.
4. **How does existing literature strengthen the current gap we are studying?** Show how previous studies provide the foundation for your research by demonstrating what has been learned, where evidence is lacking, and how your study builds on existing knowledge to generate new insights.

The Purpose of a Literature Review

1. What do we already know?

This question focuses on **established evidence**. The literature review should summarize the current state of knowledge by identifying key findings, trends, and areas of agreement across previous studies.

- ❖ Rather than listing studies one by one, authors should synthesize the evidence to show what is consistently known about the topic.

Example: *Numerous studies have shown increasing antimicrobial resistance among Escherichia coli isolates causing urinary tract infections in both hospital and community settings. High resistance to commonly prescribed antibiotics such as ampicillin and trimethoprim-sulfamethoxazole has been reported across several African countries.*

The Purpose of a Literature Review

2. What do we still not know?

Once existing knowledge has been established, the literature review should identify **what remains uncertain**.

These may be inconsistencies between studies, understudied populations, methodological limitations, or areas where evidence is lacking.

- ❖ Ask yourself:
 - Are there conflicting findings?
 - Are some populations underrepresented?
 - Is the evidence outdated?
 - Have important variables not been explored?

Example: *Although antimicrobial resistance has been widely studied, limited evidence exists on resistance patterns in county referral hospitals in Kenya, particularly following the implementation of recent antimicrobial stewardship programmes.*

The Purpose of a Literature Review

3. Why is this study necessary?

After identifying the gap, explain **why filling it matters**.

- ❖ A strong literature review should convince readers that addressing the identified gap will improve scientific understanding, inform policy, enhance clinical practice, or support future research.



Example: *Understanding current resistance patterns in county referral hospitals is essential for informing empirical antibiotic prescribing, strengthening antimicrobial stewardship programmes, and improving patient outcomes.*

The Purpose of a Literature Review

4. How does existing literature strengthen the current gap we are studying?

Previous studies provide the **foundation** upon which the current study is built.

Literature review demonstrates that the problem is important, show what has already been discovered, and justify why further investigation is needed.

- **Existing studies** provide the foundation.
- **The identified gap** is the missing knowledge
- **Your study** completes that missing section by contributing new evidence.

Example: *Previous studies consistently demonstrate increasing antimicrobial resistance in East Africa and identify E. coli as the predominant uropathogen. However, most studies have been conducted in tertiary hospitals or are more than five years old. Building on this evidence, the current study examines recent resistance patterns in county referral hospitals, providing updated data to guide treatment recommendations and antimicrobial stewardship efforts*

Characteristics of Strong Literature Reviews

A strong literature review must critically synthesize and....

- **Organize evidence logically** – Present studies in a coherent structure, e.g by theme, methodology, population, or chronology
- **Compare findings across studies** – Examine areas of agreement and disagreement, highlighting patterns and trends in the existing evidence.
- **Identify inconsistencies** – Point out conflicting findings, unanswered questions, or variations in results that warrant further investigation.
- **Highlight methodological limitations** – Critically assess weaknesses in previous studies, e.g small sample sizes, study design, that affect the strength of the evidence.
- **Establish research gaps** – Clearly demonstrate where knowledge is lacking and explain how these gaps justify the need for the current study.
- **Lead naturally to the study objectives** – Conclude the review by showing how the identified gap informs the research question, objectives, or hypotheses.

Types of literature review methods

Review Method	What It Means	When to Use It	Example of How Findings Are Presented
Narrative (Traditional) Review	Summarizes and critically discusses existing literature to provide background and justify a study.	When introducing a research topic and identifying research gaps.	<i>Several studies have consistently reported increasing antimicrobial resistance among urinary tract pathogens in sub-Saharan Africa, particularly E. coli, suggesting a growing public health concern.</i>
Systematic Review	Uses a structured, transparent process to identify and synthesize all relevant evidence addressing a specific research question.	When answering a focused research question using the best available evidence.	<i>Across the 18 studies included in this review, antimicrobial stewardship programmes were consistently associated with reduced inappropriate antibiotic prescribing and lower resistance rates.</i>
Scoping Review	Maps the breadth and nature of existing evidence and identifies research gaps.	When exploring an emerging or complex topic or assessing the available evidence.	<i>The review identified 76 studies on antimicrobial resistance in East Africa; however, most focused on tertiary hospitals, with limited evidence from primary healthcare facilities.</i>
Integrative Review	Synthesizes evidence from quantitative, qualitative, and mixed-methods studies to provide a comprehensive understanding of a topic.	When integrating different types of evidence to understand a complex issue.	<i>Quantitative studies demonstrated rising resistance rates, while qualitative evidence highlighted inappropriate antibiotic use and weak stewardship programmes as key contributing factors.</i>

Types of literature review methods - Continued

Review Method	What It Means	When to Use It	Example of How Findings Are Presented
Methodological Review	Examines and compares the research methods, measurement tools, or analytical approaches used across studies.	When evaluating the strengths and weaknesses of research methodologies.	<i>Most studies employed disc diffusion techniques based on CLSI guidelines, although considerable variation existed in sampling strategies and resistance reporting methods.</i>
Meta-analysis	Statistically combines quantitative findings from multiple studies to estimate an overall effect.	When comparable quantitative studies are available for statistical pooling.	<i>The pooled analysis demonstrated that antimicrobial stewardship interventions reduced inappropriate antibiotic prescribing by 28% (RR = 0.72; 95% CI: 0.65–0.80).</i>
Rapid Review	Uses streamlined systematic review methods to generate evidence within a shorter timeframe.	When timely evidence is needed to inform policy or programme decisions.	<i>Evidence from 12 rapidly reviewed studies suggests that implementing prescribing guidelines can reduce unnecessary antibiotic use in hospital settings, although further high-quality studies are needed.</i>

Common Mistake	Why It Is a Problem	How to Improve It
Summarizing instead of synthesizing	Simply describes each study without showing relationships, patterns, or differences.	Compare and contrast findings across studies and discuss overall trends.
Poor organization	Presents studies in a random order, making the review difficult to follow.	Organize the review by themes, methodology, population, or chronology.
Using outdated or irrelevant literature	May not reflect current evidence and weakens the justification for the study.	Prioritize recent, high-quality, and relevant peer-reviewed studies.
Lack of critical analysis	Reports findings without evaluating the strengths, weaknesses, or quality of the evidence.	Critically appraise study designs, sample sizes, methods, and limitations.
Ignoring conflicting evidence	Creates a biased review by presenting only studies that support the author's argument.	Discuss both consistent and conflicting findings and explain possible reasons for differences.
Weak or missing research gap	Readers cannot see what is unknown or why the study is necessary.	Clearly identify unanswered questions, inconsistencies, or limitations in the existing literature.
Poor use of citations	Overuses direct quotations, cites too few sources, or relies heavily on one author.	Paraphrase effectively and support arguments with multiple credible sources.
No link to study objectives	The review ends abruptly without leading to the research question or objectives.	Conclude by explaining how the literature supports and justifies the current study.
Describing every study in detail	Makes the review lengthy and repetitive without adding value.	Focus on the most relevant findings and synthesize studies with similar conclusions.
Using low-quality sources	Reduces the credibility and reliability of the literature review.	Use peer-reviewed journal articles, reputable reports, and authoritative academic sources.

Structure of a Comprehensive Literature Review (Thesis/Dissertation)

Section	Purpose	Key Questions Answered	Examples
1. Introduction Background and Problem	Introduce the topic, provide background information and context. State the problem clearly and explain how the chapter is organized.	<i>What is the issue and why is it important?</i>	Adolescent depression is a growing public health concern worldwide. In sub-Saharan Africa, the problem is compounded by limited access to mental health services... This chapter reviews the literature related to depression among adolescents.
2. Theoretical Framework	Present the theory or theories that explain the phenomenon under study and show how they guide your research.	<i>Which theory explains the phenomenon?</i>	<i>This study is guided by the Health Belief Model, which explains health behaviors based on perceived susceptibility, severity, benefits, and barriers.</i>
3. Empirical Literature Review (Organized by Objectives/Themes)	Review and critically analyze previous studies relevant to your variables or objectives. Organize them into themes, not by author or year.	<i>What do previous studies tell us?</i>	Studies have examined knowledge, attitude, and access to health services as predictors of contraceptive use. For example, Oduro (2020) found that knowledge significantly influences use, while Ali (2021) reported no association...
4. Conceptual Framework	Present the relationships among the key variables in your study, usually using a diagram and accompanying explanation.	<i>How are the variables expected to relate?</i>	<i>Figure 2.1 presents the conceptual framework showing that knowledge influences attitude, which in turn affects contraceptive uptake, with self-efficacy as a moderating variable.</i>
5. Summary and Research Gap	Summarize the reviewed literature, identify the gap in knowledge, restate the problem, and explain how your study will address the gap.	<i>What is still unknown and why is this study needed?</i>	Although many studies have examined these factors, few have focused on adolescents in rural settings in Kenya. Therefore, this study seeks to fill this gap by examining...

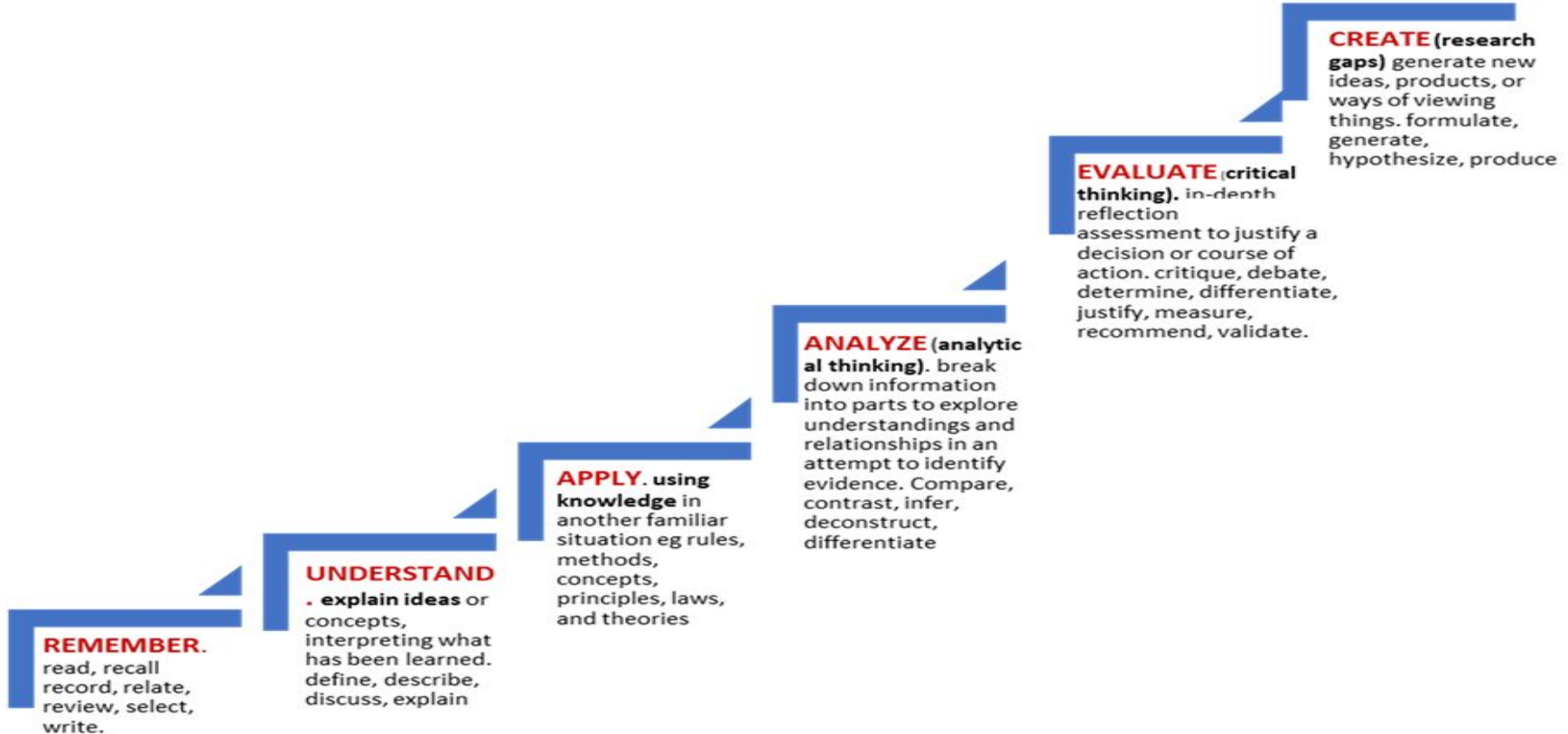
► **Outcome:** This review leads to the purpose of the study and the research objectives/questions.

Synthesizing the Review into Manuscript Paragraphs

A manuscript still contains all the elements of a literature review, but they are integrated into the Introduction rather than presented as separate headings.

Step	What to Do	Sentence Starter
1. Start with the main message.	Begin with the overall finding or theme.	<i>Several studies have shown that...</i>
2. Group similar findings together.	Discuss studies that report similar results in one sentence rather than separately.	<i>Studies conducted in Kenya, Uganda and Tanzania consistently reported...</i>
3. Compare or contrast the evidence.	Highlight agreements, differences or inconsistencies across studies.	<i>However, other studies found... / In contrast... / Similarly...</i>
4. Explain what the evidence means.	Briefly interpret the collective findings without going into detailed discussion.	<i>Together, these findings suggest that...</i>
5. End with the gap.	Identify what remains unknown and link it to your study.	<i>However, little is known about...</i>

Reading, writing and stages of learning



Constructing Scientific Arguments

Every section of a scientific manuscript contributes to one overarching argument:

A **scientific argument** is a logical, evidence-based explanation that persuades readers that:

- your research is important,
- your methods are appropriate,
- your findings are trustworthy,
- your conclusions are supported by the evidence.

Unlike personal opinions, **scientific arguments are built on data, existing literature, and sound reasoning.**

A strong scientific argument is built by connecting evidence to conclusions through logical reasoning.

Constructing Scientific Arguments

A well-constructed scientific argument should convince readers that:

- **The problem matters** – Demonstrate that the research addresses an important scientific, clinical, or societal issue by highlighting its significance and supporting it with existing evidence.
- **The methods are appropriate** – Explain why the study design, data collection procedures, and analytical methods are suitable for answering the research question and producing reliable results.
- **The findings are credible** – Present results accurately and objectively, supported by robust data, appropriate statistical analyses, and transparent reporting.
- **The conclusions are justified** – Draw conclusions that are directly supported by the findings, acknowledge study limitations, and avoid making claims that go beyond the available evidence.

Elements of a Strong Scientific Argument

- Authors must explain **what the evidence shows, what it means, and why it matters**.
Four key elements contribute to a persuasive scientific argument.

Element	What It Is	Question It Answers	Example
Claim	The main point, conclusion, or position that the author wants the reader to accept. It forms the central message of the argument.	What am I trying to prove or demonstrate?	<i>School-based interventions improve adolescent mental health.</i>
Evidence	The facts, data, statistics, observations, or research findings that support the claim. Evidence should be accurate, relevant, and credible.	What evidence supports my claim?	<i>Students receiving the Shamiri Intervention had significantly lower depression scores than those in the control group ($\beta = -2.4$, $p < 0.01$).</i>
Interpretation	The explanation of what the evidence means and how it answers the research question. It connects the results to the broader context without simply repeating the findings.	What do these findings tell us?	<i>These findings suggest that positive psychology interventions can effectively reduce depressive symptoms among adolescents in school settings.</i>
Significance	The explanation of why the findings matter and how they contribute to research, policy, practice, or future studies. It highlights the broader implications of the study.	Why are these findings important?	<i>The findings support integrating school-based mental health programmes into national adolescent health strategies to improve adolescent wellbeing.</i>

Building Logical Flow in Academic Writing

A well-written scientific manuscript presents ideas in a logical sequence that guides readers from one point to the next.

Each paragraph should focus on

- **One central idea**, beginning with a **topic sentence** that introduces the main claim
- This should be followed by **supporting evidence**, such as data, research findings, or literature that substantiates the claim.
- The author should then provide an **interpretation**, explaining what the evidence means and how it contributes to answering the research question.
- Finally, the paragraph should end with a **transition** that links the discussion to the next idea, creating a smooth and coherent flow throughout the manuscript.

Golden Rule: *One paragraph should communicate one main idea: introduce it, support it with evidence, explain its meaning, and connect it to the next idea.*

THE 5-STEP FLOW FOR A STRONG LITERATURE REVIEW PARAGRAPH

STEP	WHAT TO DO	SENTENCE STARTERS	EXAMPLE
1. TOPIC SENTENCE	Start with the main idea or overall finding of the theme.	• Several studies have shown that... • Evidence suggests that... • Previous research indicates...	<i>Several studies have shown that adolescent contraceptive uptake remains low in many low- and middle-income countries.</i>
2. EVIDENCE FROM MULTIPLE STUDIES	Provide supporting evidence from more than one study.	• For example... • Studies by... found that... • Research in [setting] reported...	<i>For example, Smith (2020), Ali (2021), and Otieno (2022) found that stigma, limited access to youth-friendly services, and inadequate knowledge are major barriers.</i>
3. COMPARISON (AGREEMENT/ DISAGREEMENT)	Compare findings across studies. Are they consistent or different?	• However... • In contrast... • Conversely... • While most studies agree..., some... / others...	<i>While most studies agree on the importance of access and stigma, some (e.g., Wang, 2023) reported a weaker association between knowledge and contraceptive uptake.</i>
4. SYNTHESIS (WHAT DO THE STUDIES COLLECTIVELY MEAN?)	Interpret the evidence together. What is the collective message?	• Taken together... • Collectively, these studies suggest... • Overall, the evidence indicates...	<i>Taken together, these studies suggest that both individual factors (knowledge, attitudes) and health system factors (access, youth-friendly services, stigma) influence adolescents' use of contraception.</i>
5. KNOWLEDGE GAP / TRANSITION	Identify what is still unknown and transition to your study.	• However, little is known about... • Despite this evidence... • Few studies have examined...	<i>However, limited evidence exists on these factors among adolescents in rural Kenya. Therefore, this study seeks to examine...</i>

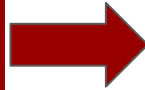
KEY TAKEAWAY

One paragraph, one message. Start with the message, support it with evidence from multiple studies, compare the findings, synthesize what they mean, and end by identifying what remains unknown or smoothly transitioning to the next theme.

Questions to Ask While Writing Every Paragraph

Poor Synthesis (Listing Studies)

Smith (2020) found that contraceptive uptake was low among adolescents. Ali (2021) reported that stigma affected contraceptive use. Otieno (2022) found that access to youth-friendly services was limited. Wang (2023) found that knowledge influenced contraceptive uptake.



Instead of asking "What did Author XY say?", ask:

1. What is the main message of this paragraph?
2. Which studies support this message?
3. Do the studies agree or disagree?
4. What does the collective evidence tell us?
5. What is still missing?



Good Synthesis (Journal Style)

Several studies have shown that contraceptive uptake among adolescents remains low despite increasing awareness of modern family planning methods. Across these studies, stigma, limited access to youth-friendly services and inadequate reproductive health information emerged as the most common barriers to contraceptive use. Although knowledge was generally associated with higher contraceptive uptake, the strength of this relationship varied across settings. Together, these findings suggest that both individual and health system factors influence adolescents' access to contraception. However, limited evidence exists on these factors among adolescents in rural Kenya, highlighting the need for context-specific research

How to avoiding weak arguments

Weak arguments reduce the credibility of a manuscript and make it difficult for readers, reviewers, and editors to trust the conclusions. Authors should avoid:

- **Unsupported opinions** – Making claims without providing evidence from the study or existing literature. Every assertion should be supported by credible data/literature
- **Overgeneralization** – Drawing broad conclusions that extend beyond the study population, setting, or findings. Conclusions should remain consistent with scope.
- **Emotional language** – Using subjective or persuasive words such as *remarkable*, *alarming*, *obviously*, or *unfortunately*. Scientific writing should remain objective,
- **Logical jumps** – Moving from evidence to conclusions without explaining the reasoning that connects them.
- **Unsupported Speculations** – Making assumptions or proposing explanations that are not supported by the study findings. Distinguish between evidence-based conclusions and speculation.

Writing an Effective Methods Section

The Methods section explains **how the study was conducted** so that readers can understand, evaluate, and, where appropriate, replicate the research.

A well-written Methods section should be **clear, precise, transparent, and sufficiently detailed**, allowing readers to assess the validity and reliability of the study without unnecessary or excessive information.

Remember: *The Methods section answers the question: "**How was the study conducted?**"*

Principle 1 – Write Clearly and Sequentially

Describe the study procedures in the exact order they were carried out. A logical sequence helps readers follow the research process from beginning to end and makes the study easier to understand and replicate.

Rather than jumping between different aspects of the methodology, present the procedures as they occurred during the study.

Example: Logical sequence in the Methods section



Tip: Write the Methods section as though you are guiding another researcher through the study step by step.

Principle 2 – Be Specific and Precise

Use clear and specific language to describe what was done.

Avoid vague statements that leave readers guessing about the procedures or methods used.

The goal is to provide enough detail for someone else to understand and replicate

Poor

Samples were analysed using standard laboratory methods.

Better

Urine samples were cultured on CLED agar, and antimicrobial susceptibility testing was performed using the Kirby–Bauer disk diffusion method according to CLSI guidelines.

Principle 3 – Write Concisely

The Methods section should include all the information necessary to understand and replicate but avoid unnecessary detail that don't contribute to the scientific understanding of the study.

Too Detailed

The questionnaires were printed on white A4 paper and distributed by trained research assistants using blue pens during the morning sessions.

Concise

Trained research assistants administered structured questionnaires to participants during scheduled data collection sessions.

Principle 4 – Use Consistent Scientific Language

Scientific writing should be objective, consistent, and precise.

The Methods section is typically written in the **past tense**, because it describes procedures that have already been completed. Use standard scientific terminology and avoiding informal or conversational language.

 **Poor** - We kind of selected participants from different hospitals.

 **Better** - Participants were recruited from four county referral hospitals using consecutive sampling.

 **Poor** - Blood was tested in the lab.

 **Better** - Blood samples were analysed using...

Principle 5 – Ensure Transparency and Reproducibility

A strong Methods section provides enough information for readers to judge the quality of the research and, if necessary, repeat the study. Be transparent about the procedures used, including any adaptations, limitations, or established guidelines followed.

For example, clearly state:

- Sampling approach
- Data collection tools
- Laboratory or analytical methods
- Statistical software
- Reporting guidelines (where applicable)

Remember: Transparency builds confidence in your findings and allows other researchers to verify or build upon your work.

Writing an Effective Results Section

The Results section communicates the findings of the study in a **clear, logical, and objective** manner.

Its purpose is to report **what was found**, allowing readers to understand the key outcomes of the research without interpretation or personal opinion.

A well-written Results section should present findings systematically, making it easy for readers to follow the progression of the study and determine whether the research objectives have been achieved.

Remember: *The Results section answers the question: **What did the study find?***

Rules for Writing Strong Results

- ✓ Follow the order of your study objectives.
- ✓ Lead with the most important finding.
- ✓ Report precise numerical results.
- ✓ Highlight key messages instead of repeating tables.
- ✓ Present findings objectively and reserve interpretation for the Discussion.

The Results section should allow readers to answer one question with confidence: **What did the study find?**

Principle 1 – Present Findings in the Order of Your Objectives

- Organize your Results section so that the findings follow the same sequence as your research objectives or research questions.
- This creates a logical flow that helps readers easily connect each objective with its corresponding result.
- Jumping between unrelated findings can make the section confusing and difficult to follow.

Tip: Readers should never have to search for the results of a particular objective.

Principle 2 – Begin with the Main Finding

Start each paragraph with the **most important finding**, followed by supporting details or statistics.

Leading with the key result immediately tells readers what was discovered and helps them focus on the most important information.

Poor

Several analyses were conducted to determine antimicrobial resistance among bacterial isolates.

Better

Antimicrobial resistance was detected in **68% of bacterial isolates**, with the highest resistance observed for ampicillin.

Principle 3 – Report Numbers Clearly

Scientific findings should be supported by **precise numerical data** rather than vague descriptions.

Clear reporting allows readers to judge the magnitude and significance of the findings.

Instead of saying:

 **Resistance was high.**

Write:

 **Resistance to ampicillin was 74.6%, compared with 18.3% for nitrofurantoin ($p < 0.001$).**

Tip: Replace subjective descriptions with measurable evidence.

Principle 4 – Highlight, Don't Repeat Tables

Tables and figures are designed to present detailed data efficiently.

The accompanying text should **highlight the key findings**, not repeat every value already shown.

Poor

The text repeats every number presented in Table 2.

Better

As shown in **Table 2**, resistance was highest for ampicillin and lowest for nitrofurantoin, indicating considerable variation in antibiotic effectiveness.

Tip: Use the text to explain the story behind the table, not to duplicate it.

Principle 5 – Remain Objective

The Results section should report **facts**, not opinions or interpretations.

Describe what the data show using neutral, evidence-based language.

Poor

This worrying finding demonstrates poor prescribing practices.

Better

Resistance was significantly higher among patients previously exposed to antibiotics (**p = 0.003**).

Remember: *Report the findings in the Results section. Explain their meaning in the Discussion section.*

Questions

Take-Home Exercise

1. Draft an abstract from their own research.
2. Draft a the last paragraph of a possible introduction summarizing the problem. Gap, and how your study addressed this gap.
3. Draft one paragraph words describing the methods.

Present work in 3 PPT slides